

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Advanced Math

12

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Q1**D**D. $4x^{-1}y^4$

Choice D is correct. Taking the square root of an exponential expression halves the exponent, so

$\frac{\sqrt{16x^4y^8}}{x^3} = \frac{4x^2y^4}{x^3}$, which further reduces to $\frac{4y^4}{x}$. This can be rewritten as $4x^{-1}y^4$.

Choice A is incorrect and may result from neglecting the denominator of the given expression and from incorrectly calculating the square root of 16. Choice B is incorrect and may result from rewriting $\frac{1}{x}$ as x^1 rather than x^{-1} . Choice C is incorrect and may result from taking the square root of the variables in the numerator twice instead of once.

Q2

B

B. 14

Choice B is correct. Let x be the first integer and let y be the second integer. If the first integer is 11 greater than twice the second integer, then $x = 2y + 11$. If the product of the two integers is 546, then $xy = 546$. Substituting $2y + 11$ for x in this equation results in $2y + 11y = 546$. Distributing the y to both terms in the parentheses results in $2y^2 + 11y = 546$. Subtracting 546 from both sides of this equation results in $2y^2 + 11y - 546 = 0$. The left-hand side of this equation can be factored by finding two values whose product is -546 , or $-1,092$, and whose sum is 11. The two values whose product is $-1,092$ and whose sum is 11 are 39 and -28 . Thus, the equation $2y^2 + 11y - 546 = 0$ can be rewritten as $2y^2 + 28y - 39y - 546 = 0$, which is equivalent to $2yy - 14 + 39y - 14 = 0$, or $2y + 39y - 14 = 0$. By the zero product property, it follows that $2y + 39 = 0$ and $y - 14 = 0$. Subtracting 39 from both sides of the equation $2y + 39 = 0$ yields $2y = -39$. Dividing both sides of this equation by 2 yields $y = -\frac{39}{2}$. Since y is a positive integer, the value of y is not $-\frac{39}{2}$. Adding 14 to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting 14 for y in the equation $xy = 546$ yields $x14 = 546$. Dividing both sides of this equation by 14 results in $x = 39$. Therefore, the two integers are 14 and 39, so the smaller of the two integers is 14.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the larger of the two integers.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**A****A. 1**

Choice A is correct. Applying the distributive property on the left- and right-hand sides of the given equation yields $x^2 + 2x + 3x + 6 = x^2 - 2x - 3x + 6 + 10$, or $x^2 + 5x + 6 = x^2 - 5x + 16$.

Subtracting x^2 from and adding $5x$ to both sides of this equation yields $10x + 6 = 16$.

Subtracting 6 from both sides of this equation and then dividing both sides by 10 yields $x = 1$.

Choices B, C, and D are incorrect. Substituting 0, -2 , or -5 for x in the given equation will result in a false statement. If $x = 0$, the given equation becomes $6 = 16$; if $x = -2$, the given equation becomes $0 = 30$; and if $x = -5$, the given equation becomes $6 = 66$. Therefore, the values 0, -2 , and -5 aren't solutions to the given equation.

Q4**C****C. 1 day after the new product launched, the company's estimated stock price is \$ 83.**

Choice C is correct. In the given equation, x represents the number of days after a new product launched, where $0 \leq x \leq 20$, and y represents the estimated stock price, in dollars, for a certain company. Therefore, the best interpretation of $x, y = 1, 83$ in this context is that 1 day after the new product launched, the company's estimated stock price is \$ 83.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q5**D**

D. $n = \frac{16}{p-20}$

Choice D is correct. To express n in terms of p , the given equation must be solved for n . Subtracting 9 from both sides of the given equation yields $p - 9 = \frac{14}{n}$. Since n is not equal to 0, multiplying both sides of this equation by n yields $p - 9n = 14$. It's given that $p > 9$, which means $p - 9$ is not equal to 0. Therefore, dividing both sides of $p - 9n = 14$ by $p - 9$ yields $\frac{p - 9n}{p - 9} = \frac{14}{p - 9}$, or $n = \frac{14}{p - 9}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q6

B

B. $f(x) = (x - 24)(x - 30) - 8$

Choice B is correct. According to the table, $f(24) = -8$, $f(30) = -8$, and $f(32) = 8$. Let the quadratic function g be defined by $g(x) = f(x) + 8$. Therefore, $g(24) = f(24) + 8$, $g(30) = f(30) + 8$, and $g(32) = f(32) + 8$. Substituting -8 for $f(24)$ in the equation $g(24) = f(24) + 8$ yields $g(24) = -8 + 8$, or $g(24) = 0$. Substituting -8 for $f(30)$ in the equation $g(30) = f(30) + 8$ yields $g(30) = -8 + 8$, or $g(30) = 0$. Substituting 8 for $f(32)$ in the equation $g(32) = f(32) + 8$ yields $g(32) = 8 + 8$, or $g(32) = 16$. By the factor theorem, if $g(24) = 0$ and $g(30) = 0$, then $g(x) = k(x - 24)(x - 30)$, where k is a constant. Since $g(32) = 16$, substituting 32 for x and 16 for $g(x)$ in the equation $g(x) = k(x - 24)(x - 30)$ yields $16 = k(32 - 24)(32 - 30)$, or $16 = k(8)(2)$, which is equivalent to $16 = 16k$. Dividing both sides of this equation by 16 yields $1 = k$. Substituting 1 for k in the equation $g(x) = k(x - 24)(x - 30)$ yields $g(x) = (x - 24)(x - 30)$. Since $g(x) = f(x) + 8$, it follows that $g(x) - 8 = f(x)$. Subtracting 8 from both sides of the equation $g(x) = (x - 24)(x - 30)$ yields $g(x) - 8 = (x - 24)(x - 30) - 8$. Substituting $f(x)$ for $g(x) - 8$ in this equation yields $f(x) = (x - 24)(x - 30) - 8$.

Choice A is incorrect. If this equation defined f , then when $x = 24$, the value of $f(x)$ would be 4 , not -8 .

Choice C is incorrect. If this equation defined f , then when $x = 32$, the value of $f(x)$ would be 32 , not 8 .

Choice D is incorrect. If this equation defined f , then when $x = 32$, the value of $f(x)$ would be -32 , not 8 .

Q7**either 7**

The correct answer is either 7, 8, or 13. Since the given equation has two integer solutions, the expression on the left-hand side of this equation can be factored as $(x+c)(x+d)$, where c and d are also integers. The product of c and d must equal the constant term of the original quadratic expression, which is 12. Additionally, the sum of c and d must be a negative number since it's given that $a > 0$, but the sign preceding a in the given equation is negative. The possible pairs of values for c and d that satisfy both of these conditions are -4 and -3 , -6 and -2 , and -12 and -1 . Since the value of $-a$ is the sum of c and d , the possible values of $-a$ are $-4 + (-3) = -7$, $-6 + (-2) = -8$, and $-12 + (-1) = -13$. It follows that the possible values of a are 7, 8, and 13. Note that 7, 8, and 13 are examples of ways to enter a correct answer.

Q8**15000**

The correct answer is 15,000. Let y represent the population of Smithville x years after 1659. Since the population doubled every 75 years, the relationship between x and y can be represented by an exponential equation in the form $y = ab^{\frac{x}{k}}$, where a is the population of Smithville in 1659 and the population increases by a factor of b every k years. It's given that the population doubled, or increased by a factor of 2, every 75 years. Therefore, $b = 2$ and $k = 75$. Substituting 2 for b and 75 for k in the equation $y = ab^{\frac{x}{k}}$ yields $y = a2^{\frac{x}{75}}$. It's also given that the population of this town was 240,000 in 1959. The number of years from 1659 to 1959 is $1959 - 1659$, or 300 years. Therefore, when $x = 300$, the value of y is 240,000. Substituting 300 for x and 240,000 for y in the equation $y = a2^{\frac{x}{75}}$ yields $240,000 = a2^{\frac{300}{75}}$, or $240,000 = a2^4$, which is equivalent to $240,000 = a(16)$. Dividing both sides of this equation by 16 yields $15,000 = a$. Therefore, the population of this town in 1659 was 15,000.

Q9**-9**

The correct answer is -9. Since w is in the denominator of a fraction in the given equation, w can't be equal to 0. Since w isn't equal to 0, multiplying both sides of the given equation by w yields an equivalent equation, $-54 = 6w$. Dividing both sides of this equation by 6 yields $-9 = w$. Therefore, -9 is the solution to the given equation.

Q10

B

B. $gx = \frac{a-19}{x-4} + 2$

Choice B is correct. It's given that the graph of $y = gx$ is produced by translating the graph of $y = fx$ 3 units down and 4 units to the right in the xy -plane. Therefore, function g can be defined by an equation in the form $gx = fx - 4 - 3$. Function f is defined by the equation $fx = \frac{a-19}{x} + 5$, where a is a constant. Substituting $x - 4$ for x in the equation $fx = \frac{a-19}{x} + 5$ yields $fx - 4 = \frac{a-19}{x-4} + 5$. Substituting $\frac{a-19}{x-4} + 5$ for $fx - 4$ in the equation $gx = fx - 4 - 3$ yields $gx = \frac{a-19}{x-4} + 5 - 3$, or $gx = \frac{a-19}{x-4} + 2$. Therefore, the equation that defines function g is $gx = \frac{a-19}{x-4} + 2$.

Choice A is incorrect. This equation defines a function whose graph is produced by translating the graph of $y = fx$ 3 units down and 4 units to the left, not 3 units down and 4 units to the right.

Choice C is incorrect. This equation defines a function whose graph is produced by translating the graph of $y = fx$ 4 units to the left, not 3 units down and 4 units to the right.

Choice D is incorrect. This equation defines a function whose graph is produced by translating the graph of $y = fx$ 4 units to the right, not 3 units down and 4 units to the right.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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